A clean route to zeta regularization:

heat kernel, Mellin transform, and meromorphic continuation

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Cleaned working note

Abstract

This note separates three distinct operations that are often conflated in informal discussions of zeta regularization: ordinary summation, regulator-dependent finite parts, and analytic continuation. The basic example is the exponential cutoff of $\sum_{n\geq 1} n$, followed by the Mellin-transform construction of the Riemann zeta function and its spectral generalization for Laplace-type operators on compact manifolds.

1 Three notions that must be kept separate

There are three different objects in play.

- An *ordinary sum* $\sum_n a_n$ means the limit of partial sums, when that limit exists.
- A *regularization* replaces a divergent expression by a convergent family $F(\varepsilon)$, studies the asymptotics as $\varepsilon \to 0^+$, and may extract a finite part.
- An *analytic continuation* extends a holomorphic function from its initial domain to a larger domain, typically as a meromorphic function.

The slogan $1 + 2 + 3 + \cdots = -1/12$ is therefore misleading unless it is explicitly interpreted as a statement about a particular regularized finite part or about the analytically continued value of the Riemann zeta function at $s = -1$.

2 A convergent cutoff calculation

For $\varepsilon > 0$, define

$$S(\varepsilon) = \sum_{n=1}^{\infty} n e^{-\varepsilon n}. \tag{1}$$

This is an ordinary convergent sum for each positive ε. With $q = e^{-\varepsilon}$,

$$\sum_{n\geq 1} n q^n = \frac{q}{(1-q)^2}, \tag{2}$$

so

$$S(\varepsilon) = \frac{e^{-\varepsilon}}{(1-e^{-\varepsilon})^2}. \tag{3}$$

The Bernoulli expansion gives

$$\frac{\varepsilon}{1-e^{-\varepsilon}} = 1 + \frac{\varepsilon}{2} + \frac{\varepsilon^2}{12} - \frac{\varepsilon^4}{720} + O(\varepsilon^6), \tag{4}$$

and hence

$$S(\varepsilon) = \frac{1}{\varepsilon^2} - \frac{1}{12} + O(\varepsilon^2) \qquad (\varepsilon \to 0^+). \tag{5}$$

The divergent part is ε^{-2} and the cutoff-independent constant term is $-1/12$.

Finite part in this scheme. For this exponential cutoff one may define

$$\sum_{n\geq 1}^{(\mathrm{FP})} n := \mathrm{FP}_{\varepsilon\to 0^+} \sum_{n=1}^{\infty} n e^{-\varepsilon n} = -\frac{1}{12}, \tag{6}$$

where FP means the constant term in the small-ε asymptotic expansion after divergent powers have been removed. This is not an ordinary sum.

3 Mellin transforms as the systematic mechanism

For a function $F(t)$ on $(0,\infty)$, its Mellin transform is

$$\mathcal{M}[F](s) = \int_0^\infty t^{s-1} F(t)\,\mathrm{d}t, \tag{7}$$

where the integral converges. The important structural fact is that small-t asymptotics of $F(t)$ control the pole structure of its Mellin transform.

Suppose that as $t \to 0^+$,

$$F(t) \sim \sum_j c_j t^{\alpha_j}. \tag{8}$$

Then the terms $c_j t^{\alpha_j}$ generate possible simple poles at $s = -\alpha_j$ in the Mellin transform. The standard continuation procedure is to subtract finitely many singular small-t terms, integrate the subtracted expression in a larger strip, and then add back the subtracted terms as explicit meromorphic functions of s.

4 Riemann zeta from a heat-kernel type integral

For $\mathrm{Re}(s) > 1$,

$$\zeta(s) = \sum_{n=1}^{\infty} n^{-s}. \tag{9}$$

Using

$$\frac{1}{e^t - 1} = \sum_{n=1}^{\infty} e^{-nt} \tag{10}$$

and

$$\int_0^\infty t^{s-1} e^{-nt}\,\mathrm{d}t = \Gamma(s) n^{-s}, \tag{11}$$

one obtains, again for $\mathrm{Re}(s) > 1$,

$$\zeta(s) = \frac{1}{\Gamma(s)} \int_0^\infty \frac{t^{s-1}}{e^t - 1}\,\mathrm{d}t. \tag{12}$$

Near $t = 0$,

$$\frac{1}{e^t - 1} = \frac{1}{t} - \frac{1}{2} + \frac{t}{12} - \frac{t^3}{720} + \cdots . \tag{13}$$

Subtracting finitely many terms of this expansion and adding them back explicitly produces the meromorphic continuation of $\zeta(s)$. In that analytically continued sense,

$$\zeta(-1) = -\frac{1}{12}. \tag{14}$$

This value agrees with the finite part of the exponential cutoff above, but the two statements should still be distinguished conceptually.

5 Spectral zeta functions and heat traces

Let M be a compact Riemannian manifold and let $\Delta \geq 0$ be the Laplace–Beltrami operator. Its spectrum is discrete,

$$0 = \lambda_0 \leq \lambda_1 \leq \lambda_2 \leq \cdots, \qquad \lambda_k \to \infty. \tag{15}$$

The heat trace is the convergent quantity

$$\Theta(t) = \mathrm{Tr}(e^{-t\Delta}) = \sum_{k \geq 0} e^{-t\lambda_k}, \qquad t > 0. \tag{16}$$

As $t \to 0^+$ it has an asymptotic expansion

$$\Theta(t) \sim (4\pi t)^{-\dim M/2} \sum_{j \geq 0} a_j t^{j/2}. \tag{17}$$

The spectral zeta function is initially defined for $\mathrm{Re}(s)$ large by

$$\zeta_\Delta(s) = \sum_{k \geq 1} \lambda_k^{-s}. \tag{18}$$

Removing the zero mode from the heat trace gives the Mellin representation

$$\zeta_\Delta(s) = \frac{1}{\Gamma(s)} \int_0^\infty t^{s-1} \big(\Theta(t) - 1\big)\,\mathrm{d}t, \tag{19}$$

where this integral converges. The heat-kernel expansion at small t then generates the meromorphic continuation of $\zeta_\Delta(s)$ to the complex plane. The coefficients a_j determine the possible pole structure.

6 A good sentence and a bad sentence

A careful statement is:

> In the exponential cutoff scheme, the finite part of $\sum_{n\geq 1} ne^{-\varepsilon n}$ as $\varepsilon \to 0^+$ is $-1/12$; equivalently, the analytically continued Riemann zeta function satisfies $\zeta(-1) = -1/12$.

A bad statement is:

> $1 + 2 + 3 + \cdots = -1/12.$

The second phrase erases the distinction between ordinary summation, regularization, and analytic continuation.